

# lec 1.4

24 September 2026

20:30

## DBSCAN

### 1. What is DBSCAN?

**DBSCAN (Density-Based Spatial Clustering of Applications with Noise)** is an **Unsupervised Machine Learning Algorithm** that groups together data points that are **closely packed (high-density regions)** and identifies isolated points as **noise (outliers)**.

Unlike K-Means:

- DBSCAN **does not require the number of clusters (K)** before training.
- It can discover **clusters of arbitrary (irregular) shapes**.
- It can automatically detect **noise/outliers**.

### Full Form

DBSCAN  
=  
Density-Based Spatial Clustering of  
Applications with Noise

### Main Idea

↓  
Dataset

↓  
Find Dense Regions

↓  
Expand Clusters

Expand Clusters

Separate Sparse Regions

Detect Noise Points

## 2. Why was DBSCAN Developed?

Traditional clustering algorithms like K-Means have several limitations.

- ✗ Must specify K beforehand
- ✗ Cannot detect noise
- ✗ Works best for spherical clusters
- ✗ Sensitive to outliers

DBSCAN solves these problems by using **density** instead of centroids.

